

SHADOW AI DATA GUARD

A Simple Guide for Your Cyber Security Team Project

1. What is "Shadow AI" Really?

Imagine a "Shadow" following you—it sees everything you do but stays hidden. **Shadow AI** is when employees or students use AI tools (like ChatGPT) to finish work faster, but they do it **without permission** from the security team.

The Real-World Example:

A bank employee is tired. He asks ChatGPT: *"Hey, fix the errors in this list of customers: [John Doe, Account #445566, Balance \$50,000]."*

As soon as he hits 'Enter', that private bank data is no longer inside the bank. It is now on the AI company's servers. **This is a massive security leak.**

2. Why is it Important?

In 2026, companies aren't losing data to hackers as much as they are **giving it away** to AI models.

- **Privacy:** If a hospital leaks patient data via AI, they can be sued for millions.
- **Secrecy:** If a tech company's secret code is pasted into an AI, their competitors might find it later.
- **Grade Factor:** Your Mam will love this because it solves a **2026 problem**, not a 2010 one!

3. How Efficient should it be?

A security tool shouldn't make the computer slow.

- It must check the text **instantly** (Latency $L < 0.5s$).
- It must be **Accurate**. If it hides normal words, users will get annoyed and turn it off.

Problem Statement for your Report:

"With 75% of workers using AI tools for daily tasks, organizations face 'Shadow AI' data exfiltration risks. This project develops a **Semantic Data Guardrail** using Python-based NLP to detect, flag, and redact Personally Identifiable Information (PII) from AI prompts before they leave the local network."

4. Team Roles (Easy Start)

Member 1: Data Scout

Find a list of common "Sensitive Entities" (Emails, Names, Credit Card patterns).

Member 2: The Logic Brain

Write a Python script using the library '**SpaCy**' to find these entities in a sentence.

Member 3: The Masker

Write the code that replaces "John" with "USER_X" and "1234-5678" with "[HIDDEN_CARD]".

Member 4: The Interface

Build a simple website using '**Streamlit**' where you type a prompt and it shows the "Cleaned" version.

5. Reference Papers (Search these for your report)

- *"Semantic Data Loss Prevention in the Age of Generative AI"* (IEEE 2025).
- *"The Rise of Shadow AI: Risk and Mitigation Strategies"* (ResearchGate 2026).
- *"PII-Redact: A Lightweight Model for Secure AI Prompting"* (arXiv 2025).